INFOMDW: Data Wrangling

Algorithmic Fairness

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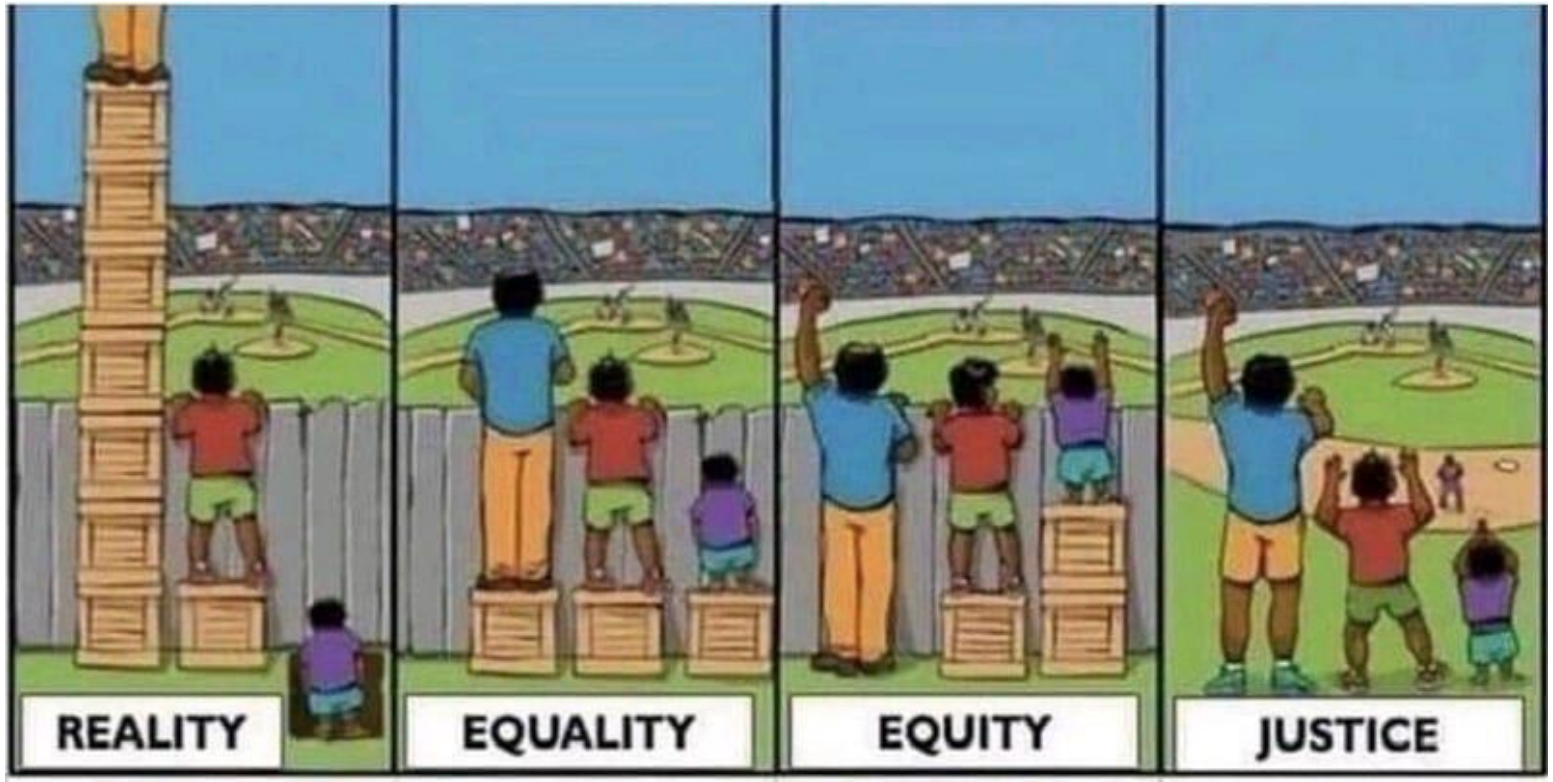
Topics for Today

- Algorithmic Fairness
  - Fairness Measures
  - Bias Mitigation Algorithms
  - Fairness in other ML models

Reading Material & Exercises

- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. ACM computing surveys (CSUR), 54(6), 1-35

What is Fairness?



REALITY	EQUALITY	EQUITY	JUSTICE
One gets more than is needed, while the other gets less than is needed. Thus, a huge disparity is created.	The assumption is that everyone benefits from the same supports. This is considered to be equal treatment.	Everyone gets the support they need, which produces equity.	All 3 can see the game without supports or accommodations because the cause(s) of the inequity was addressed. The systemic barrier has been removed.

What is Fairness?

"Fairness is the absence of any prejudice or favoritism towards an individual or a group based on their intrinsic or acquired traits in the decision-making context"

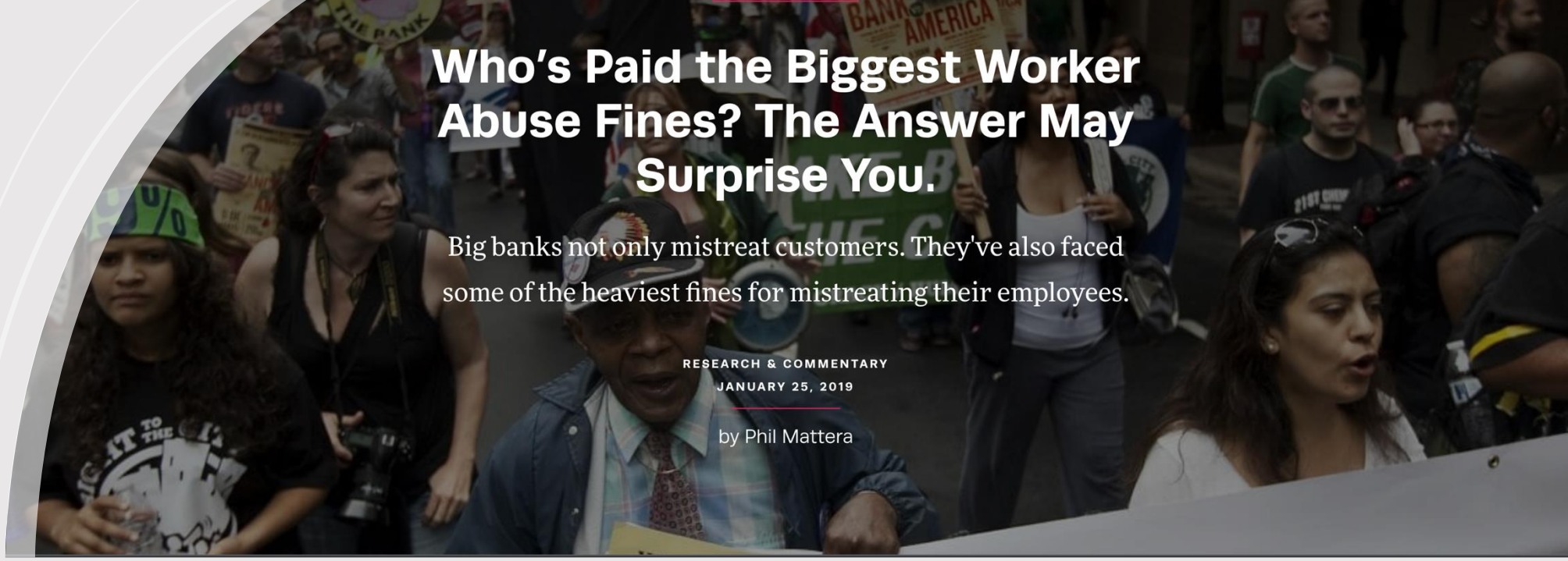
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Bank of America fined $335m for minority discrimination

21 December 2011

https://www.bbc.com/news/business-16296146



Who's Paid the Biggest Worker Abuse Fines? The Answer May Surprise You.

Big banks not only mistreat customers. They've also faced some of the heaviest fines for mistreating their employees.

RESEARCH & COMMENTARY

JANUARY 25, 2019

by Phil Mattera

Parent Companies with the Highest Disclosed Discrimination Penalties

Rank	Parent	Penalty Total	Cases
1	Bank of America	$210,296,593	8
2	Coca-Cola	$200,616,000	9
3	Novartis	$183,000,000	2
4	Morgan Stanley	$150,385,000	6
5	Abercrombie & Fitch	$90,115,600	4
6	FedEx	$80,035,138	15
7	Boeing	$79,935,059	7
8	Verizon Communications	$71,504,891	6
9	Wells Fargo	$68,099,000	5
10	SoftBank (parent of Sprint)	$62,852,756	3
11	Walmart	$52,888,031	27

https://inequality.org/research/penalties-workplace-abuse/

Protected Attributes



What are protected characteristics?

It is against the law to discriminate against someone because of:

- age
- disability
- gender reassignment
- marriage and civil partnership
- pregnancy and maternity
- race
- religion or belief
- sex
- sexual orientation

These are called protected characteristics.

Related content

Equality Act 2010

Advice and guidance

ADVICE AND GUIDANCE

Age discrimination

ADVICE AND GUIDANCE

Disability discrimination

https://www.equalityhumanrights.com/equality/equality-act-2010/protected-characteristics

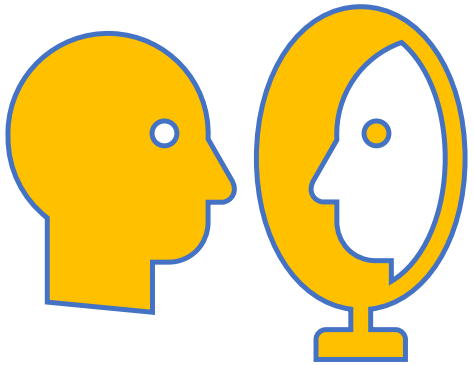
Fairness Measures

- Demographic groups are determined based on sensitive attribute
  - Also called protected attributes
- Privileged and unprivileged groups are determined based on the sensitive attributes and the decision label
  - Groups that receive undesirable decision more frequently are unprivileged
- Checking parity between the demographic groups
- Cannot always identify hidden unfairness



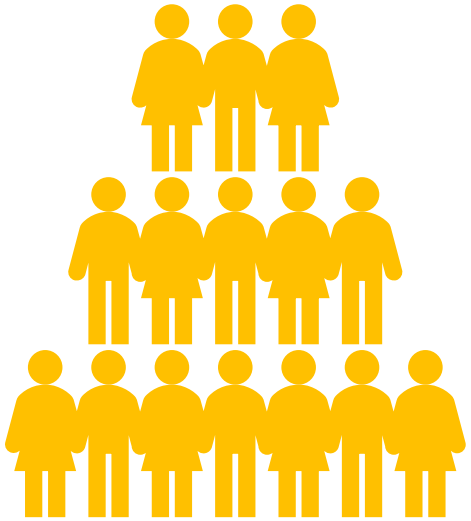
Individual Fairness Measures

- Individuals with similar features except the sensitive (protected) attributes must have the same/similar outcomes
- A similarity/distance measure is needed
- Requires strong assumptions regarding the relationship between features (variables) and the decision label



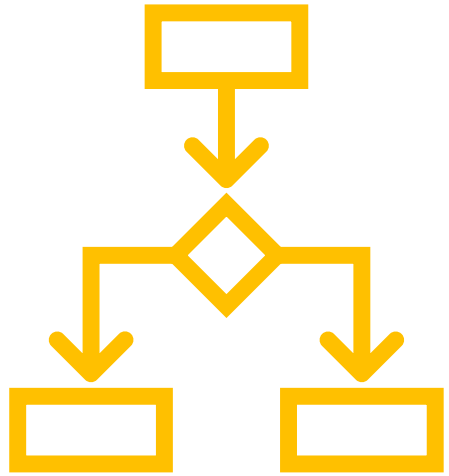
Group Fairness Measures

- Define multiple subgroups in a dataset
- Check parity between these subgroups
- A statistical constraint is needed:
  - E.g.: false positive rates



Causal Fairness Measures

- Causal relationships between the attributes and the outcome labels
- Sensitive attributes should not affect the outcome labels
- Identify “proxy” attributes
- Constructing the correct causal graph is a must
- Mostly a domain expert is needed



Fairness Measures in Details

- $D = \{X, S, Y\}$ is a dataset,
- X: the set of attributes that does not contain sensitive information regarding individuals
- S: is the set of sensitive attributes containing sensitive information
- $Y/\hat{Y} \in \{0,1\}$: the original/predicted class label of individuals, which indicates the decision outcome
- G/G': the values of unprivileged/privileged group

Fairness Measures – Demographic Parity (DP) Difference

- The instances in both unprivileged and privileged groups should have equal probability to receive positive outcomes

$$DP_{diff} = P\left[Y(\mathbf{x}) = 1 \mid S(\mathbf{x}) = G'\right] - P\left[Y(\mathbf{x}) = 1 \mid S(\mathbf{x}) = G\right] \approx 0.$$

- This measure takes values between 0 and 1 where 0 is the optimal

Fairness Measures – Disparate Impact (DI) Ratio

- The ratio between the probability of privileged and unprivileged groups getting positive or desired outcomes

$$DI(D) = \frac{P\left[Y(\mathbf{x}) = 1 | S(\mathbf{x}) = G\right]}{P\left[Y(\mathbf{x}) = 1 | S(\mathbf{x}) = G'\right]}.$$

- A dataset or a classifier is considered fair (by law) if its DI-ratio is between 0.8 and 1.25 (1 is the optimal)

Fairness Measures – Consistency

- An individual fairness measure determines how similar the labels are for the similar instances in a dataset based on the k-neighbors of the instance

$$Consistency = 1 - \frac{1}{|D|}\sum_{i=1}^{|D|}\left|\widehat{y}(\mathbf{x}_i) - \frac{1}{|kNN(\mathbf{x}_i)|}\sum_{\mathbf{x}_j \in kNN(\mathbf{x}_i)}\widehat{y}(\mathbf{x}_j)\right|$$

- This measure: takes values between 0 and 1 with 1 is the optimal
- **NOTE:** DP-diff., DI-ratio and consistency can be computed from the original dataset and the outcomes of a ML model

Fairness Measures – Equalized Odds (EO) Difference

- EO states that instances from privileged and unprivileged groups should have equal True Positive Rate (TPR) and False Positive Rate (FPR)

$$P_1 = P\left[\widehat{Y}(\mathbf{x}) = 1 \mid S(\mathbf{x}) = G', Y(\mathbf{x}) = 1\right],$$
$$P_2 = P\left[\widehat{Y}(\mathbf{x}) = 1 \mid S(\mathbf{x}) = G, Y(\mathbf{x}) = 1\right],$$
$$P_3 = P\left[\widehat{Y}(\mathbf{x}) = 1 \mid S(\mathbf{x}) = G', Y(\mathbf{x}) = 0\right],$$
$$P_4 = P\left[\widehat{Y}(\mathbf{x}) = 1 \mid S(\mathbf{x}) = G, Y(\mathbf{x}) = 0\right].$$

- For a classifier to be fair:

$$P_1 = P_2 \text{ and } P_3 = P_4$$

- i.e.

$$AEO_{diff} = \frac{(P_1 - P_2) + (P_3 - P_4)}{2}$$

Fairness Measures – Predictive Parity (PP)

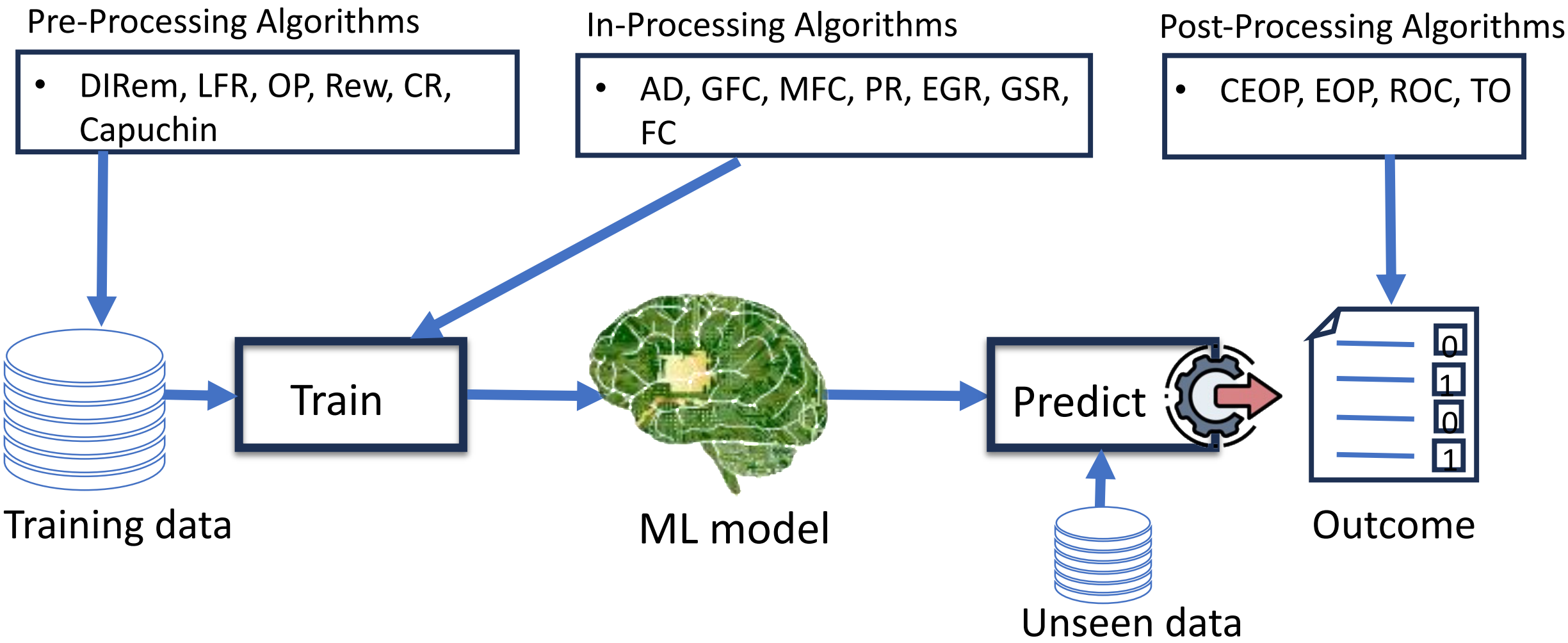
- A classifier is fair in terms of predictive parity if the prob. that an example is positive in the original dataset given that it is predicted positive from both privileged and unprivileged groups is the same

$$P\left[Y(\mathbf{x}) = 1 \mid \widehat{Y}(\mathbf{x}) = 1, S(\mathbf{x}) = G\right] = P\left[Y(\mathbf{x}) = 1 \mid \widehat{Y}(\mathbf{x}) = 1, S(\mathbf{x}) = G'\right]$$

- This measure and the AEO measure can be applied on the outcome of an ML only (cannot be computed from the original dataset)

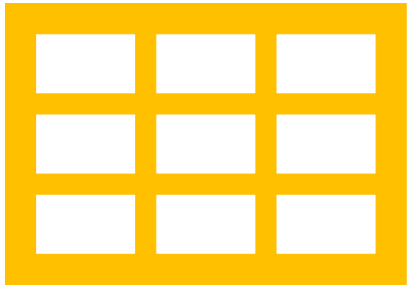
Bias and Fairness

Mitigation Algorithms


Pre-Processing Algorithms
• DIRem, LFR, OP, Rew, CR, Capuchin
In-Processing Algorithms
• AD, GFC, MFC, PR, EGR, GSR, FC
Post-Processing Algorithms
• CEOP, EOP, ROC, TO
Train
Predict
0
1
0
1
Training data
ML model
Outcome
Unseen data

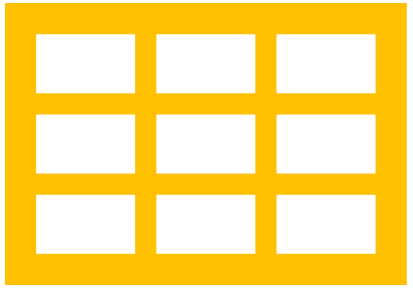
Mitigation Algorithms – Pre-Processing Techniques

- Pre-process the dataset only
- Different types of strategies:
  - Fairness through “unawareness”: deletes the sensitive attributes in a dataset
  - Preferential sampling (re-sampling): data objects are sampled with replacement
  - Massaging (Relabeling): changes the actual class label of some of the instances in the training set
  - Reweighing: assigns weights to each instance in the training set



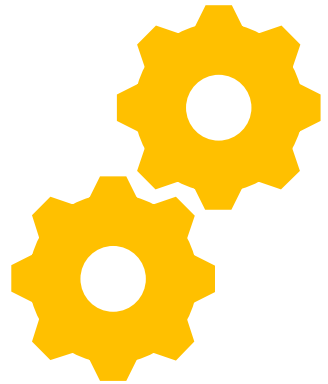
Mitigation Algorithms – Pre-Processing Techniques

- Example
- Learning Fair Representations (LFR) [ref]:
  - Intermediate representation of training set
  - Accurate representation but conceals information about sensitive attributes
  - In the final mapping, the class labels are changed to satisfy group and individual fairness



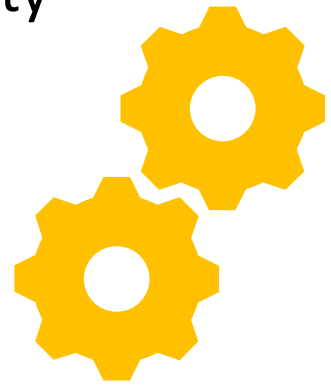
Mitigation Algorithms – In-Processing Techniques

- Adjust/tune the classification algorithm
- Applied during the model training
- Regularization of the model
- Dependent on the implemented classifier



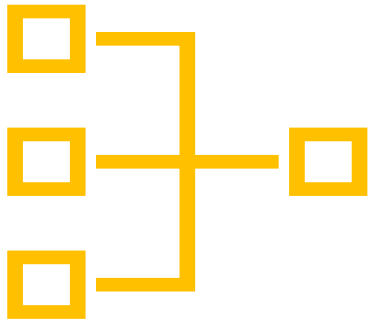
Mitigation Algorithms – In-Processing Techniques

- Example
- Adversarial Debiasing (Adv. Deb.): In-processing [ref]
  - An adversarial learning technique, one predictor and one adversary
  - Predictor predicts the class label, adversary predicts the sensitive attribute
  - Must satisfy demographic parity, equalized odds and equal opportunity



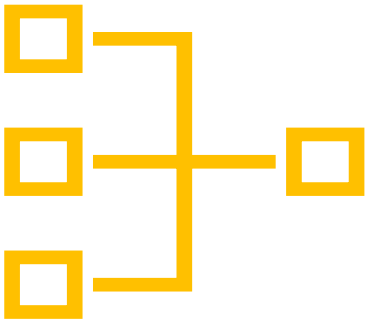
Mitigation Algorithms – Post-Processing Techniques

- Eliminate the discrimination from the final predictions
- Change the predicted outcomes of classifiers
- Changes are based on certain rules or constraints such as equalized odds.
- Thresholding the critical regions
  - E.g., change the samples that are predicted positive from the privileged group with prob. < 0.6 to negative.



Mitigation Algorithms – Post-Processing Techniques

- Example
- Calibrated Equalized Odds (Calibrated EO): Post Processing [ref]
  - Targets to satisfy both calibration and equalized odds
  - To achieve the goal, it changes some of the predicted class labels



Bias Detection Tools

- Only detect/quantify the amount of bias
- Audit focus
- Some of the developed tools:
  - FairML
  - Themis
  - What-If Tool (Google)

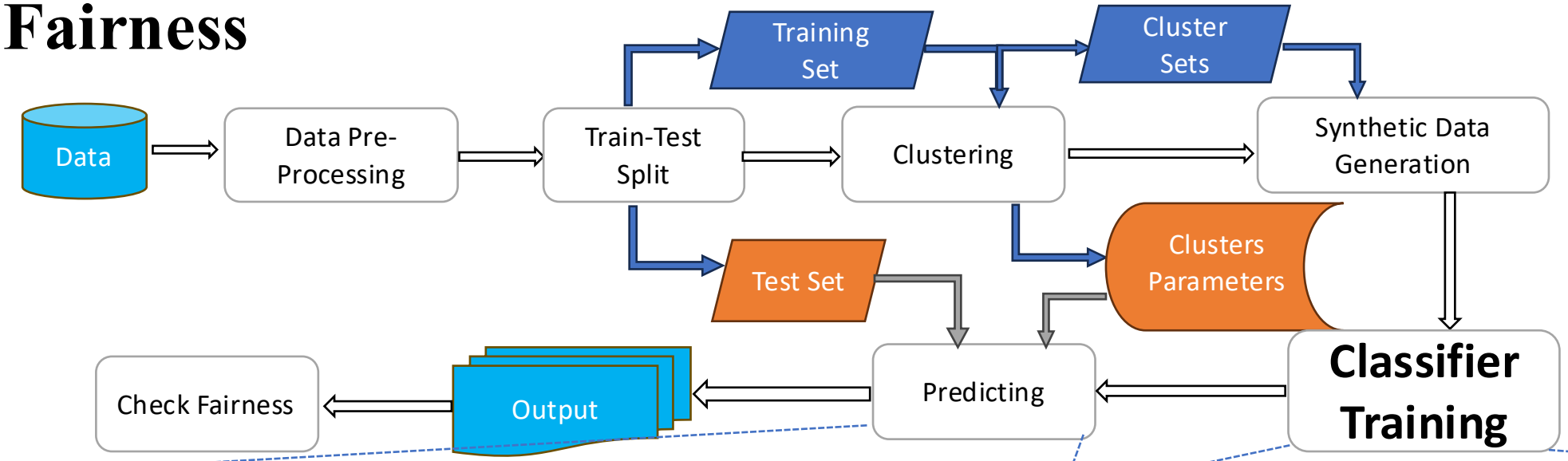


Bias Mitigation Tools

- Detect / quantify the amount of bias AND eliminate or mitigate it
- Internal / business focus
- Some of the developed tools:
  - Themis-ML
  - FairLearn (Microsoft)
  - AI Fairness 360 (IBM)



SynthFair: Synthetic Data Generation for Ensuring Subgroup Fairness



One model: $\hat{y} = m(x)$

Str.1

$\hat{y} = \omega_1 m_1(x) + \cdots + \omega_k m_k(x)$

Str.2

Test Example(x)

$\hat{y} = \omega_i m_i(x)$,
$where\ x \in Cluster\ (C_i)$

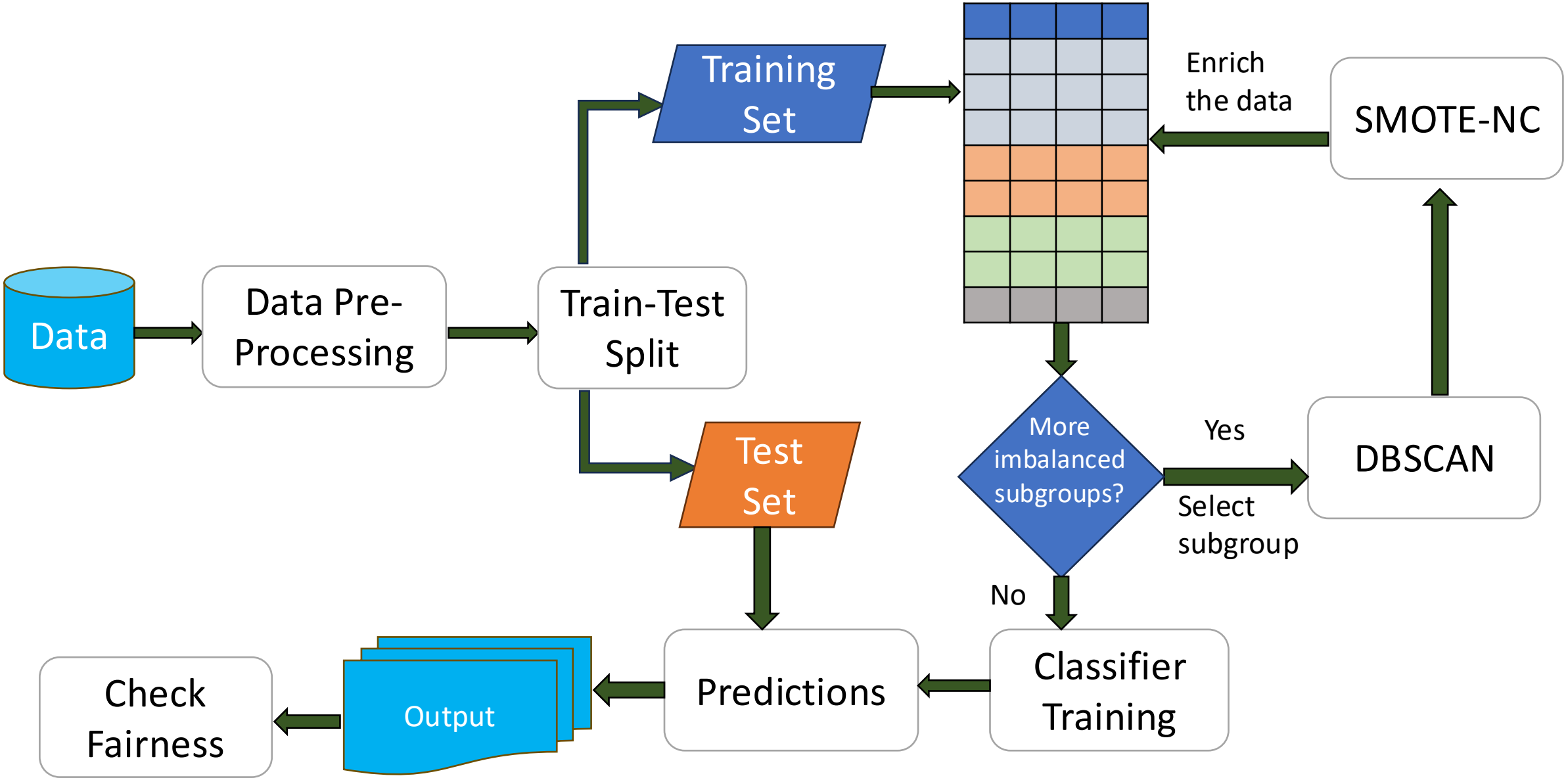
Str.3

Train one classifier (m)

Train k classifiers $(m_1, m_2, \ldots, m_k)$

Cluster Sets with Synthetic data

OSFFR: OverSampling For Fair Representation



Non–Traditional Fairness Problems (Regression)

- Classification use threshold on outcomes of a continuous variable
  - Success in a job (hiring)
  - GPA of the first year (scholarship)
  - Risk assessment factor (loan/credit approval)
  - …

Non–Traditional Fairness Problems (Regression)

- Training/testing sets contain three types of attributes
  - $\boldsymbol{x}$ set of non-sensitive attributes
  - $\boldsymbol{s}$ set of sensitive attributes (such as gender, age, religion, etc)
  - $\boldsymbol{y}$ the target variable (continuous random variable)
  - Demographic groups ($\Leftrightarrow$ classification)
  - Problem
    - How to define fairness in this context?

Non–Traditional Fairness Problems (Regression)

- Coefficient of Determination (R^2)
  - y is the set of observed values , $\hat{y}$ the predicted values
  - $R^2 = \frac{Var(\hat{y} - y)}{Var(y - \bar{y})}$, $\bar{y}$ = mean of the observed values
- Fair regression
  - Consider $R^2 = \frac{Var(\hat{y} - \tilde{y})}{Var(\hat{y})}$,
    - $\tilde{y}$ = predicted values using only non-sensitive attributes

Non–Traditional Fairness Problems (Regression)

- Minimizing the expected loss
  - y is the set of observed values , $\hat{y}$ the predicted values
  - $\ell(y, \hat{y})$, could be MAE, MSE, etc.
  - $minimize\ \mathbb{E}\left[(\ell(y, \hat{y})|\boldsymbol{s} = G_p) - (\ell(y, \hat{y})|\boldsymbol{s} = G_u)\right]$

Non–Traditional Fairness Problems (PCA)

- Minimizing the reconstruction error
  - M is a matrix ∈ $R^{m \times n}$, transform M into $\widehat{M} \in R^{m \times n}$ with rank ($d < n$)
  - $minimize \left\| M - \widehat{M} \right\|_F$
- Fair PCA
  - Two populations privileged group G_p, and the unprivileged group G_u
  - Minimize $\left\| G_p - \hat{G}_p \right\|_F - \left\| G_u - \hat{G}_u \right\|_F$

Non–Traditional Fairness Problems (Clustering)

- Clustering can be viewed as unsupervised classification
  - Partition the students such that low GPA students are in one cluster
  - Customers with a risk of default on the loan in one cluster
- Fair clustering
  - Restricted dominance: at most m individual from G_p in each cluster
  - Minority protection: at least m' individual from G_u in each cluster



Wrap-Up

- We discussed
  - Bias and Imbalanced Data
  - Fairness measures
  - Bias Mitigation Algorithms
  - Fairness in different ML models

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